

KCCF School Clinic System - Complete Documentation

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Client: Kurios Christian Colleges Foundation

Client Location: Magallanes, Cavite

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Project Overview

KCCF School Clinic System is a comprehensive web-based health management system designed for Kurios Christian Colleges Foundation. It streamlines patient triage, health record management, medicine inventory tracking, and clinic operations.

Key Objectives

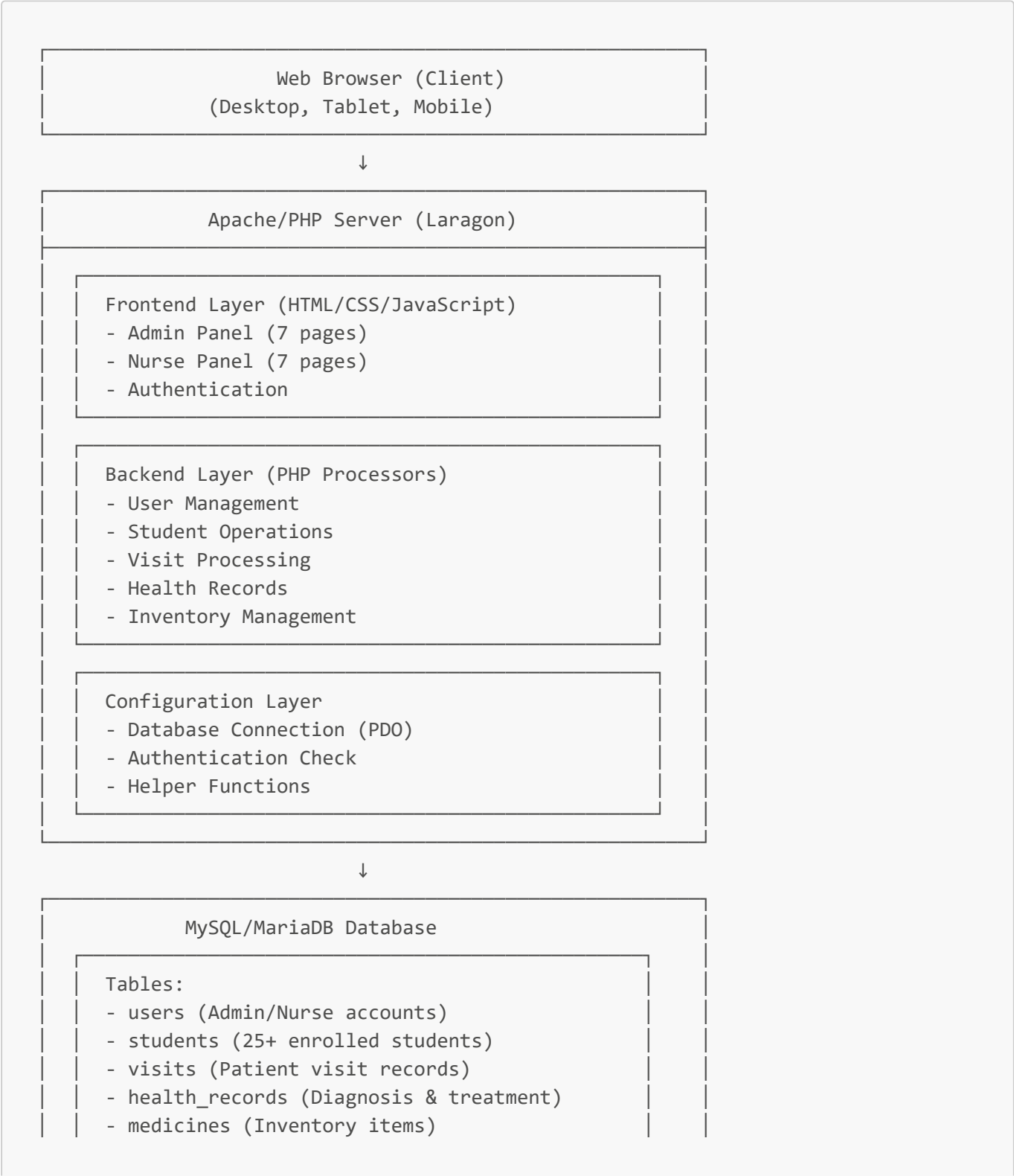
- Centralized student health record management
- Real-time patient queue management
- Medicine inventory tracking with low-stock alerts
- Role-based access control (Admin & Nurse)
- Comprehensive clinic reporting and analytics
- Mobile-responsive design for accessibility

Project Scope

- **Students:** 25+ enrolled students across 4 programs
- **Programs:** BSED, BSIT, BSCRIM, BSBA (Years 1-4 each)
- **Users:** Admin (1), Nurses (2+)
- **Records:** 60+ historical visits, comprehensive health data

System Architecture

High-Level Architecture



Technology Stack

Component	Technology	Version
Server	Apache HTTP Server	2.4+
Runtime	PHP	7.4+
Database	MySQL/MariaDB	5.7+
Frontend	HTML5, CSS3, JavaScript (ES6+)	
Frameworks	PDO (Database Abstraction)	PHP Native
Authentication	bcrypt (Password Hashing)	PHP Native
UI Library	Phosphor Icons	v2.0+
PDF Export	html2pdf.js	v0.10.1
Development	XAMPP (Local Environment)	

Installation & Setup

Prerequisites

- XAMPP (Apache + MySQL + PHP) installed
- PHP 7.4 or higher
- MySQL 5.7 or higher
- Modern web browser (Chrome, Firefox, Safari, Edge)

Step 1: XAMPP Installation & Configuration

1. Download XAMPP:

- Visit: <https://www.apachefriends.org/>
- Download version with PHP 7.4 or higher
- Install to default location (C:\xampp)

2. Start Services:

- Open XAMPP Control Panel
- Click "Start" for Apache
- Click "Start" for MySQL
- Verify both show green "Running"

3. Access phpMyAdmin:

- Open browser and go to: <http://localhost/phpmyadmin>

- Default credentials: Username: `root`, Password: (empty)

Step 2: Database Setup

1. In phpMyAdmin:

- Click "New" to create new database
- Database name: `school_clinic_system`
- Collation: `utf8_general_ci`
- Click "Create"

2. Import Database Schema:

- Click on `school_clinic_system` database
- Go to "Import" tab
- Click "Choose File" and select SQL script or copy schema from `database_structure.txt`
- Click "Go" to execute

3. Import Seed Data:

- Copy-paste SQL content from `seed.txt`
- Execute in SQL query window
- Verify 25 students, 12 medicines, 60+ visits created

Step 3: Project File Placement

1. Quick clone into XAMPP htdocs (recommended):

- Open PowerShell (or Command Prompt) and run:

```
git clone https://github.com/aldrieBQRN/school-clinic-system.git
"C:\xampp\htdocs\school-clinic-system"
```

- This clones the repository directly into the webroot so files are immediately accessible at `http://localhost/school-clinic-system/`.

2. Manual placement (alternative):

- Locate XAMPP htdocs: `C:\xampp\htdocs\`
- Extract or move project files to: `C:\xampp\htdocs\school-clinic-system\`

3. Verify Structure:

- `C:\xampp\htdocs\school-clinic-system\index.php` ← Login page
- `C:\xampp\htdocs\school-clinic-system\admin\` ← Admin pages
- `C:\xampp\htdocs\school-clinic-system\nurse\` ← Nurse pages
- `C:\xampp\htdocs\school-clinic-system\config\` ← Configuration

Step 4: Configuration

1. Edit Database Connection:

- Open: `config/db_connect.php`

- Verify settings:

```
$host = 'localhost';      // XAMPP MySQL host
$db = 'school_clinic_system'; // Database name
$user = 'root';           // Default XAMPP MySQL user
$pass = '';               // Default XAMPP MySQL password (empty)
$charset = 'utf8mb4';
```

2. Test Connection:

- Access: <http://localhost/school-clinic-system/>
- If database error appears, check phpMyAdmin connection first

Step 5: Verify Installation

1. Access Login Page:

- Open browser: <http://localhost/school-clinic-system/>
- Should see login form (not blank page or error)

2. Test Login:

- Use admin credentials (see next section)
- Should redirect to admin dashboard

3. Check Database:

- In phpMyAdmin, verify tables:
 - [users](#) (3 records)
 - [students](#) (25 records)
 - [visits](#) (60+ records)
 - [health_records](#), [medicines](#)

Step 6: Default Login Credentials

Admin Account:

- Username: [admin](#)
- Password: [password123](#) (bcrypt hashed)

Nurse Account 1:

- Username: [nurse_reyes](#)
- Password: [password123](#)

Nurse Account 2:

- Username: [nurse_mendoza](#)
- Password: [password123](#)

 **Security Note:** Change all passwords after first login!

Step 7: Accessing the Application

1. Ensure Services Running:

- XAMPP Control Panel shows Apache & MySQL in green

2. Open Application:

- Browser: <http://localhost/school-clinic-system/>

3. Login:

- Use credentials from Step 6
- Admin sees admin dashboard
- Nurse sees nurse dashboard

Troubleshooting XAMPP Setup

Issue	Solution
"This site can't be reached"	Verify Apache is started in XAMPP Control Panel
"Unable to connect to database"	Verify MySQL is started; check phpMyAdmin access
"Table doesn't exist"	Import seed.txt in phpMyAdmin SQL tab
"Blank page / 500 error"	Check <code>C:\xampp\apache\logs\error.log</code>
Port already in use	Stop other applications using ports 80/443 or 3306

Database Structure

Users Table

```
CREATE TABLE users (  
  user_id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(100) NOT NULL,  
  role ENUM('Admin', 'Nurse') NOT NULL,  
  status ENUM('Active', 'Inactive') DEFAULT 'Active',  
  username VARCHAR(50) UNIQUE NOT NULL,  
  password VARCHAR(255) NOT NULL,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Students Table

```
CREATE TABLE students (  
  student_id INT PRIMARY KEY AUTO_INCREMENT,  
  first_name VARCHAR(50) NOT NULL,  
  last_name VARCHAR(50) NOT NULL,
```

```
gender ENUM('Male', 'Female') NOT NULL,  
birthdate DATE NOT NULL,  
course VARCHAR(50) NOT NULL,           -- BSED, BSIT, BSCRM, BSBA  
year_level INT NOT NULL,               -- 1, 2, 3, 4  
guardian_name VARCHAR(100),  
relationship VARCHAR(50),  
guardian_contact VARCHAR(15),  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Visits Table

```
CREATE TABLE visits (  
  visit_id INT PRIMARY KEY AUTO_INCREMENT,  
  student_id INT NOT NULL,  
  complaint VARCHAR(255) NOT NULL,  
  temperature DECIMAL(5,1),           -- Celsius  
  height DECIMAL(6,2),                 -- cm  
  weight DECIMAL(6,2),                 -- kg  
  nurse_notes TEXT,  
  time_in TIME,  
  date_logged DATE,  
  status ENUM('Active', 'Completed') DEFAULT 'Active',  
  FOREIGN KEY (student_id) REFERENCES students(student_id)  
);
```

Health Records Table

```
CREATE TABLE health_records (  
  record_id INT PRIMARY KEY AUTO_INCREMENT,  
  visit_id INT NOT NULL,  
  student_id INT NOT NULL,  
  diagnosis VARCHAR(255),  
  treatment TEXT,  
  date DATETIME,  
  FOREIGN KEY (visit_id) REFERENCES visits(visit_id),  
  FOREIGN KEY (student_id) REFERENCES students(student_id)  
);
```

Medicines Table

```
CREATE TABLE medicines (  
  medicine_id INT PRIMARY KEY AUTO_INCREMENT,  
  name VARCHAR(100) NOT NULL,  
  category VARCHAR(50),  
  quantity INT DEFAULT 0,
```

```
status ENUM('In Stock', 'Low Stock', 'Out of Stock'),
expiration DATE,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

User Roles & Access

Role Hierarchy

System Administrator
(Admin Account - Full System Access)

Can:

- ✓ View all health records
- ✓ Generate reports & analytics
- ✓ Manage user accounts (Add/Edit/View)
- ✓ View complete visit logs
- ✓ Manage medicine inventory
- ✓ View student database
- ✓ Export data (PDF)

Clinical Nurse Staff
(Nurse Accounts - Clinical Operations)

Can:

- ✓ Log new patient visits
- ✓ Record vital signs
- ✓ Finalize visit diagnosis/treatment
- ✓ View health records
- ✓ Check medicine availability
- ✓ Restock medicines
- ✓ Register new students
- ✓ Edit student information

Cannot:

- X Access admin reports
- X Manage user accounts
- X View admin dashboard

Features Overview

1. Authentication & Authorization

- **Secure Login:** bcrypt password hashing
- **Session Management:** PHP sessions with timeout

- **Role-Based Access Control:** Separate admin & nurse dashboards
- **Logout:** Secure session termination

2. Dashboard

- **Admin Dashboard:** Overview of clinic statistics, user count, recent visits
- **Nurse Dashboard:** Active patient queue, medicine alerts, quick actions

3. Student Management

- **Student Registration:** Add new students with course & year level
- **Student Records:** Browse, search, edit student profiles
- **Guardian Info:** Emergency contact details management
- **Course/Year Filtering:** Organized by BSED, BSIT, BSCRIM, BSBA

4. Patient Visit Management

- **Log Visit:** Initiate new patient triage (complaint, vitals)
- **View Queue:** Real-time active patient queue
- **Finalize Visit:** Complete visit with diagnosis & treatment
- **Visit History:** Archive of all completed visits

5. Health Records

- **Record Creation:** Automatic with visit finalization
- **Record Search:** Filter by student, course, date range
- **Record Viewing:** Detailed patient medical history
- **Record Editing:** Update diagnosis and treatment info

6. Medicine Inventory

- **Add Medicines:** Stock new medicine items
- **Track Inventory:** Real-time quantity tracking
- **Stock Alerts:** Low stock and expiration warnings
- **Restock Operations:** Update quantities
- **Delete Obsolete:** Remove expired items

7. Reporting & Analytics

- **Visit Reports:** Filtered by date range, course, complaint
- **Clinic Statistics:** Patient volume, common complaints
- **Export to PDF:** Generate printable reports
- **Search & Filter:** Advanced filtering options

8. User Management

- **Add Users:** Create new admin/nurse accounts
- **Edit Users:** Modify user information and status
- **View Users:** User directory with active status
- **Password Hashing:** Secure bcrypt encryption

9. Mobile Responsiveness

- **Responsive Design:** Works on desktop, tablet, mobile
- **Mobile Menu:** Collapsible sidebar on small screens
- **Touch-Friendly:** Optimized buttons and controls
- **Adaptive Tables:** Card layout on mobile

File Structure & Descriptions

Root Level Files

index.php	# Login page and authentication entry point
logout.php	# Session termination and logout handler
database_structure.txt	# Database schema and table definitions
seed.txt	# Initial seed data for development/testing

Admin Panel (/admin/)

dashboard.php	# Admin overview with statistics and system info
health_records.php	# Search and view all student health records
inventory.php	# Medicine stock management and tracking
manage_users.php	# User account management (Add/Edit/View/Delete)
reports.php	# Generate clinic reports with filters
student_records.php	# Student database with add/edit/delete functions
visit_log.php	# Archive of completed visits (read-only)

Nurse Panel (/nurse/)

dashboard.php	# Nurse dashboard with active queue and alerts
health_records.php	# View patient health history
inventory.php	# Check medicine availability
student_records.php	# Student profiles and directory
visit_log.php	# Completed visits archive
visits.php	# Active patient queue for triage
forms/	
register_student.php	# New student registration form
finalize_visit.php	# Complete patient visit with diagnosis

Backend Processors (/backend/)

auth_login.php	# Authenticate user login
admin/	
process_add_user.php	# Create new admin/nurse account

process_edit_user.php	# Modify user details
nurse/	
process_register_student.php	# Insert new student record
process_edit_student.php	# Update student information
process_delete_student.php	# Remove student from system
process_log_visit.php	# Create new visit entry
process_finalize_visit.php	# Complete visit with diagnosis
process_edit_visit.php	# Modify visit details
process_delete_visit.php	# Remove visit record
process_delete_health_record.php	# Delete health record
process_add_medicine.php	# Add medicine to inventory
process_update_medicine.php	# Update medicine details
process_delete_medicine.php	# Remove medicine from stock
process_restock_medicine.php	# Adjust medicine quantity

Configuration (/config/)

db_connect.php	# Database connection setup (PDO)
auth_check.php	# Session verification and access control
medicine_helpers.php	# Helper functions for medicine operations

Includes (/includes/)

admin_header.php	# Admin navigation header with user menu
admin_sidebar.php	# Admin sidebar with menu options
nurse_header.php	# Nurse navigation header with notifications
nurse_sidebar.php	# Nurse sidebar with menu options
footer.php	# Common footer with copyright info

Assets (/assets/)

css/style.css	# Main stylesheet (responsive, dark theme)
js/script.js	# JavaScript utilities and mobile functionality
images/logo.jpg	# KCCF school logo

Admin Features

Dashboard

- System statistics (total students, users, recent visits)
- Quick navigation to main functions
- User greeting with current date/time

Student Records

- **Search:** By name, ID, course, gender
- **Filter:** By course and gender
- **Display:** Age calculation, registration date
- **Actions:** View profile, Edit info, Delete record
- **Export:** PDF of student directory
- **Pagination:** 10 records per page

Health Records

- **Search:** By student ID, name, or diagnosis
- **Filter:** By course and date range
- **View:** Complete medical history with diagnosis
- **Edit:** Update existing health records
- **Export:** PDF report with filters applied

Inventory Management

- **Track Stock:** Real-time quantity updates
- **Alerts:** Low stock (<5 units) and expiration warnings
- **Status:** In Stock, Low Stock, Out of Stock
- **Operations:** Add, Edit, Delete, Restock
- **Export:** Inventory report to PDF

Visit Log (Archive)

- **Completed Visits Only:** Read-only archive
- **Search:** By student name, ID, or complaint
- **Filter:** By course and date range
- **View Details:** Complete triage data and vitals
- **Statistics:** Temperature analysis with color coding

User Management

- **Add User:** Create new admin or nurse account
- **Edit User:** Modify name, role, or status
- **View All:** User directory with active status
- **Delete:** Remove inactive accounts
- **Password:** Bcrypt hashed on creation

Reports

- **Visit Analysis:** Trends and statistics
- **Date Filtering:** Custom date range selection
- **Export:** PDF generation with professional formatting
- **Charts:** Visual representation of data

Nurse Features

Dashboard

- **Active Queue:** Real-time list of patients waiting
- **Alerts:** Low medicine stock warnings
- **Quick Actions:** Buttons to register student or log visit
- **Notifications:** Medicine expiration alerts

Visits (Patient Queue)

- **Active Patients:** Real-time queue of incoming patients
- **Triage Data:** Initial complaint, vitals, notes
- **Status:** Active (In Queue) or Completed
- **Actions:** View details, Edit, Finalize visit, Delete
- **Edit:** Update complaint, temperature, height, weight, notes

Finalize Visit

- **Complete Triage:** Record final diagnosis and treatment
- **Auto-Populated:** Student and visit information pre-filled
- **Diagnosis Field:** Detailed medical assessment
- **Treatment Field:** Prescribed treatment and advice
- **Save:** Creates permanent health record

Register Student

- **New Enrollment:** Add student to system
- **Guardian Info:** Emergency contact collection
- **Course & Year:** Select from BSED, BSIT, BSCRM, BSBA (Years 1-4)
- **Validation:** Required fields enforcement
- **Confirmation:** Success notification on submission

Health Records

- **Search:** By student ID, name, course
- **Filter:** By course and date range
- **Pagination:** 10 records per page
- **View Details:** Modal with complete record
- **Vitals Display:** Temperature with color-coded status

Student Records

- **Directory:** Browse all registered students
- **Search:** By ID, name, or course
- **Filter:** By course and gender
- **View Profile:** Modal with student details
- **Edit:** Update student information
- **Delete:** Remove student (if no visit history)

Inventory

- **Medicine List:** All available medicines
 - **Stock Check:** Real-time quantity display
 - **Categories:** Organized by medicine type
 - **Alerts:** Highlighted low stock items
 - **Expiration:** Date tracking with warnings
 - **Restock:** Update quantities quickly
 - **Add Medicine:** New stock items
-

API Endpoints

Authentication

```
POST /backend/auth_login.php
  Input: username, password
  Output: Session creation / Error

GET /logout.php
  Output: Session destruction
```

Student Operations (Nurse)

```
POST /backend/nurse/process_register_student.php
  Input: first_name, last_name, gender, birthdate, course, year_level,
        guardian_name, relationship, guardian_contact
  Output: Success/Error message

POST /backend/nurse/process_edit_student.php
  Input: student_id, first_name, last_name, course, year_level,
        guardian_name, relationship, guardian_contact
  Output: Success/Error message

POST /backend/nurse/process_delete_student.php
  Input: student_id
  Output: Success/Error message
```

Visit Operations

```
POST /backend/nurse/process_log_visit.php
  Input: student_id, complaint, temperature, height, weight, nurse_notes
  Output: visit_id, Success/Error

POST /backend/nurse/process_finalize_visit.php
  Input: visit_id, diagnosis, treatment
  Output: Success/Error, Creates health_record

POST /backend/nurse/process_edit_visit.php
```

```
Input: visit_id, complaint, temperature, height, weight, nurse_notes  
Output: Success/Error
```

```
POST /backend/nurse/process_delete_visit.php  
Input: visit_id  
Output: Success/Error
```

Health Records

```
POST /backend/nurse/process_edit_health_record.php  
Input: record_id, diagnosis, treatment  
Output: Success/Error
```

```
POST /backend/nurse/process_delete_health_record.php  
Input: record_id  
Output: Success/Error
```

Medicine Operations

```
POST /backend/nurse/process_add_medicine.php  
Input: name, category, quantity, expiration  
Output: Success/Error
```

```
POST /backend/nurse/process_update_medicine.php  
Input: medicine_id, name, category, quantity, status, expiration  
Output: Success/Error
```

```
POST /backend/nurse/process_delete_medicine.php  
Input: medicine_id  
Output: Success/Error
```

```
POST /backend/nurse/process_restock_medicine.php  
Input: medicine_id, quantity  
Output: Success/Error
```

User Management (Admin)

```
POST /backend/admin/process_add_user.php  
Input: name, username, password, role  
Output: Success/Error
```

```
POST /backend/admin/process_edit_user.php  
Input: user_id, name, role, status  
Output: Success/Error
```

Security Implementation

Authentication & Authorization

- ✓ **Secure Password Storage:** bcrypt hashing (cost: 10).
- ✓ **Session Management:** PHP \$_SESSION with timeout.
- ✓ **Access Control:** Role-based routing (admin_check.php, auth_check.php).
- ✓ **CSRF Protection:** Form submissions with PHP native protection.

Data Protection

- ✓ **SQL Injection Prevention:** Prepared statements with PDO.
- ✓ **XSS Prevention:** htmlspecialchars() on all output.
- ✓ **Input Validation:** Type checking and sanitization.
- ✓ **Error Handling:** Generic error messages to users.

Database Security

- ✓ **PDO Abstraction:** Parameterized queries only.
- ✓ **Connection Isolation:** Separate user account for DB.
- ✓ **Foreign Keys:** Referential integrity enforcement.
- ✓ **Constraints:** NOT NULL, UNIQUE, DEFAULT values.

Best Practices

- ⚠ **Change Default Passwords:** After first deployment.
- ⚠ **Update PHP Version:** Keep PHP 7.4+ current.
- ⚠ **HTTPS Deployment:** Use SSL/TLS in production.
- ⚠ **Regular Backups:** Database backup schedule.
- ⚠ **Log Monitoring:** Track failed login attempts.

Troubleshooting

Common Issues

1. Database Connection Error

Error: "Error connecting to database"

Solution:

- Verify MySQL is running
- Check credentials in `config/db_connect.php`
- Ensure database `school_clinic_system` exists
- Verify user permissions

2. Login Not Working

Error: "Invalid username or password" or blank response

Solution:

- Verify user exists in `users` table
- Check password is correctly bcrypt hashed
- Ensure cookies are enabled
- Clear browser cache

3. Missing Student/Course Data

Error: "No courses available" or empty dropdowns

Solution:

- Import seed.txt into database
- Verify students table has records
- Check for NULL course values
- Refresh page cache

4. Medicine Alerts Not Showing

Error: Low stock warnings not displaying

Solution:

- Update medicine quantities manually
- Set expiration dates
- Refresh browser
- Check JavaScript console for errors

5. PDF Export Fails

Error: "PDF generation failed" or blank file

Solution:

- Verify html2pdf.js is loaded (check browser console)
- Ensure table data is present
- Try exporting smaller dataset
- Check browser security settings

6. Mobile Layout Issues

Error: Sidebar not collapsing or misaligned content

Solution:

- Clear browser cache and cookies
- Verify script.js is loaded
- Test in incognito/private window
- Check viewport meta tag in HTML

7. Year Level Display Incorrect

Error: Shows "1th Year" instead of "1st Year"

Solution:

- Verify ordinal conversion function in script.js
- Check database year_level values (1-4)
- Clear page cache
- Refresh database with seed.txt

Debug Mode

Enable error logging in `config/db_connect.php`:

```
error_reporting(E_ALL);  
ini_set('display_errors', 1);
```

Check PHP error log:

- Laragon: `C:\laragon\logs\php-error.log`
- Browser console: F12 → Console tab
- Network tab: Check failed requests

Development Notes

Recent Updates (May 2026)

Schema Migration: `grade/section` → `course/year_level`

- **Reason:** Better accommodate 4-year programs with year levels
- **Impact:** 14+ files updated, all queries refactored
- **Status:** Complete and tested

Mobile Responsiveness Improvements

- **Consolidated JS:** Moved mobile-responsive.js code to script.js
- **Responsive Tables:** Card layout on mobile (<768px)
- **Touch Optimization:** Larger buttons and spacing
- **Sidebar Collapse:** Auto-collapse on mobile

Display Formatting: Ordinal Year Levels

- **Format:** Years now display as "1st, 2nd, 3rd, 4th" instead of "1, 2, 3, 4"
- **Implementation:** `getOrdinal()` function in JavaScript and PHP
- **Applied:** All modals, tables, and displays

File Organization Cleanup

- Removed 8+ temporary documentation files
- Consolidated JavaScript utilities
- Organized project structure
- Updated documentation files

Performance Optimizations

- AJAX pagination for faster page loading
- Debounced search to reduce queries
- Modal caching to improve responsiveness
- Optimized CSS with CSS variables

Future Recommendations

1. **Authentication:**

- Implement 2FA (Two-Factor Authentication)
- Add password reset functionality
- Session security improvements

2. **Database:**

- Add audit logging for changes
- Implement soft deletes (archive records)
- Database backup automation

3. **Features:**

- SMS notifications for alerts
- Photo ID storage for students
- Prescription printing
- Appointment scheduling

4. **Performance:**

- Database indexing optimization
- API rate limiting
- Caching layer implementation
- Image optimization

5. **Security:**

- SSL/TLS certificate
- Web Application Firewall (WAF)
- Regular security audits
- Penetration testing

Support & Contact

Software Developer: John Aldrie Baquiran

Location: Nasugbu, Batangas

For Technical Support:

- Check this documentation first
- Review troubleshooting section
- Check error logs in Laragon
- Contact the software developer

Document History

Version	Date	Author	Changes
1.0	May 4, 2026	John Aldrie Baquiran (Software Developer)	Initial documentation

End of Documentation

This document is confidential and intended for authorized personnel only. Last Updated: May 4, 2026